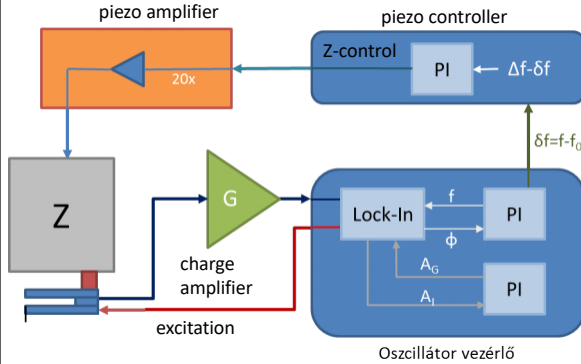


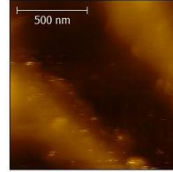
Atomic force microscope with a quartz tuning fork

The "clock signal" in the watches is provided by a tuning-fork-shaped quartz oscillator, which can be excited electrically due to its piezoelectric behaviour. By sticking an STM needle on one of the branches of the tuning fork, not only the tunnel current but also the force between the sample and the needle can be studied by the resonant frequency of the quartz oscillator. Based on the force measurement, we can even take images of insulating surfaces. To do this, a complex control scheme has to be implemented:



During scanning, the resonance frequency is varied due to the force between the needle and the sample. The PLL tells the actual resonance frequency (δf) relative to the free resonance frequency (f_0). The z-direction piezo is moved by means of a third PI controller so that the frequency detuning, i.e. the force between the needle and the sample, is constant. This corresponds to a Δf setpoint detuning. (Precisely: the frequency detuning is not sensitive to the force but to the spring constant, i.e. the force gradient)

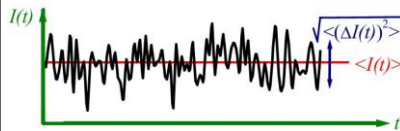
A PLL is used to excite the quartz oscillator continuously at its resonance frequency: i.e. a voltage with ω circular frequency and the current of the quartz oscillator is measured. A PI controller is used to control the frequency so that the phase shift is zero (i.e. it is excited at exactly the resonance frequency). Another PI controller is used to keep the amplitude of the oscillation constant.



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Spectral density of noise

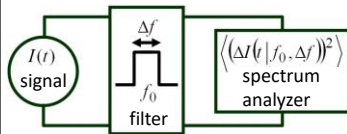
Usually, we are interested in the average value of a physical quantity (e.g. current), but around this average value we may experience fluctuations for various reasons:



The deviation of the current from the mean: $\Delta I = I(t) - \langle I(t) \rangle$

By definition: $\langle \Delta I(t) \rangle = 0$

Experimental definition of the spectral density of fluctuations: The signal $I(t)$ is passed through a band-pass filter with a narrow bandwidth Δf around a center frequency f_0 and the $\langle (\Delta I(t|f_0, \Delta f))^2 \rangle$ mean squared deviation of the such filtered signal is measured. The latter is proportional to the width of the frequency window Δf , and the proportionality factor is the $S_I(f_0)$ spectral density of the current noise at f_0 :



Spectral density of noise, or power spectrum density, PSD

Unit:

$$[S_I] = \frac{A^2}{Hz}$$

The mean squared deviation of the current over the full frequency range:

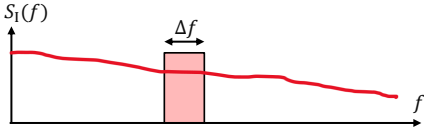
$$\langle (\Delta I(t))^2 \rangle = \int_0^\infty df S(f)$$

Similarly, for voltage noise:

$$\langle (\Delta V(t|f_0, \Delta f))^2 \rangle = S_V(f_0) \Delta f$$

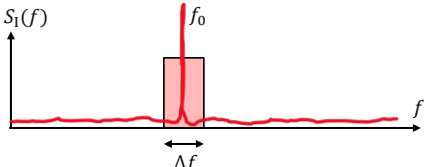
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Spectral density of noise



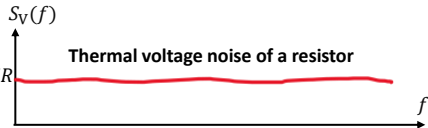
In the case of a peaked spectrum, i.e. a dominant frequency component at frequency f_0 , the total mean squared deviation within the band Δf does not depend on the width of the frequency window, only on whether f_0 is in that band or not. Therefore, a signal with a peaked spectrum is not noise, and does not even comply with the definition of noise.

Noise is a signal with continuous spectrum, with fluctuations of all frequencies, and the mean squared deviation of the various frequency components is additive. The continuous nature of the spectrum naturally yields that the overall mean squared deviation of the signal within a narrow Δf band is proportional to Δf , as included in the previous definition of the spectral density:

$$\langle (\Delta I(t|f_0, \Delta f))^2 \rangle = S_I(f_0) \Delta f$$


Thermal noise (will come later...)





If we take a simple resistor with no signal applied to it, we expect to measure zero voltage and zero current across it. In contrast, due to thermal fluctuations, we see a finite current noise when we connect the two ends of the resistor with a current meter. The spectral density of this noise is $S_I = 4kTG$, where k is the Boltzmann constant, T is the absolute temperature and $G=1/R$ is the conductance. The noise density of this so-called thermal noise is frequency independent (white noise). Similarly, if we connect a voltmeter to the resistor, we experience voltage fluctuations. These have an $S_V = S_I \cdot R^2 = 4kTR$ spectral density due to the $\Delta V = \Delta I \cdot R$ relation (Ohm's law), and $S_V \sim (\Delta V)^2$; $S_I \sim (\Delta I)^2$ (noise definition).

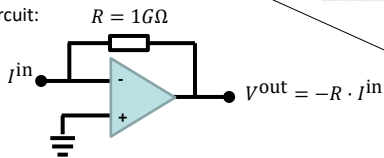
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Example: thermal noise of a current amplifier

FEMTO DLPCA-200 current amplifier used for scanning tunneling microscopy (STM) measurements.
 "10⁹" amplification (1GΩ feedback resistor),
 1nA input -> 1V output.
 Specifications:

Input noise: 4.3fA/√Hz
 Bandwidth: 1kHz

Current amplifier circuit:



Reminder: the bandwidth is related to the RC time-constant of the feedback resistor and its stray capacitance



If the input current is zero, the output is the thermal voltage noise of resistor R:

$$\langle (\Delta V^{\text{out}})^2 \rangle = S_V \cdot \Delta f; S_V = 4kTR; \Delta V_{\text{rms}}^{\text{out}} = \sqrt{4kT \cdot R \cdot \Delta f}$$

The output would have the same noise if R had no noise, but the input current would be noisy. Normalized by the bandwidth, we get the corresponding effective input noise density:

$$\Delta I_{\text{rms}}^{\text{in}} = \Delta V_{\text{rms}}^{\text{out}} / R$$

$$S_I = \langle (\Delta I^{\text{in}})^2 \rangle / \Delta f$$

$$\sqrt{S_I} = \frac{\Delta I_{\text{rms}}^{\text{in}}}{\sqrt{\Delta f}} = \sqrt{\frac{4kT}{R}} \approx 4\text{fA}/\sqrt{\text{Hz}}$$

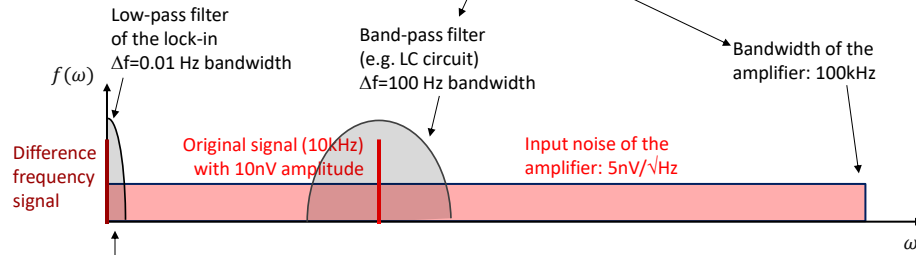
← This is the lowest noise you can get with this amplifier circuit at room temperature! For lower noise measurements, a low temperature amplifier is recommended.

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Noise filtering considerations for a low noise amplifier – why should we use a lock-in?

Calculation example: we want to measure a signal with frequency 10kHz and amplitude 10nV with a low noise voltage amplifier with input noise $\sim 5\text{nV}/\sqrt{\text{Hz}}$ (see later for the definition of noise). The amplifier has a bandwidth of 100kHz and a gain of 1000 times. The amplified amplitude of our signal is $10\mu\text{V}$, while the amplifier's input noise results in a broadband noise output of 1.6mV rms ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{100\text{kHz}} \cdot 1000$)!

A **bandpass filter** with a $Q=100$ quality factor (which is very good) can limit the bandwidth to $10\text{kHz}/Q=100\text{Hz}$. So, we see $50\mu\text{V rms}$ noise at the output of the amplifier ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{100\text{Hz}} \cdot 1000$), which is still high compared to the $10\mu\text{V}$ amplified output of the true signal.



Using the **lock-in principle** (thanks to phase-sensitive detection), we can filter out the different frequency components much better than with the bandpass filter. The bandwidth is only limited by the integration time, if we measure on the scale of 100 seconds, a filtering bandwidth of 0.01 Hz is achieved, so we only see the amplifier output noise in this narrow band, which is $0.5\mu\text{V rms}$ ($5\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{0.01\text{Hz}} \cdot 1000$), a factor of 20 less than the signal to be measured.

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Current-current correlation function

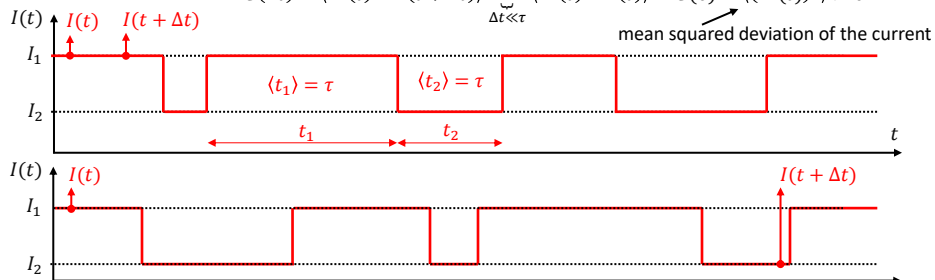
Current-current correlation function:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle = \langle \Delta I(0) \cdot \Delta I(\Delta t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \Delta I(t) \cdot \Delta I(t + \Delta t), \quad \text{Unit: } [C(\Delta t)] = A^2$$

Example: random telegraph noise (RTN). The current randomly fluctuates between two values (I_1 and I_2), such that the t_1 and t_2 times spent in the two states are random variables, but the mean value is t for both.

For $\Delta t \ll \tau$ there is a good chance, that $I(t + \Delta t) = I(t)$ (no switching), i.e.:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle \approx \langle \Delta I(t) \cdot \Delta I(t) \rangle = C(0) = \langle (\Delta I(t))^2 \rangle > 0$$



For $\Delta t \gg \tau$ the $I(t + \Delta t)$ and the $I(t)$ values are completely independent from each other (like a random number generator for coin toss), i.e.:

$$C(\Delta t) = \langle \Delta I(t) \cdot \Delta I(t + \Delta t) \rangle \approx \underbrace{\langle \Delta I(t) \rangle}_0 \cdot \underbrace{\langle \Delta I(t + \Delta t) \rangle}_0 = 0$$

The current-current correlation function is the mean squared deviation of the current for small Δt , it decays to zero for $\Delta t \rightarrow \infty$, and the transition between the two happens around the characteristic time, so-called correlation time of the system, τ in this case. This is frequently described by an exponential correlation function:

$$C(\Delta t) = \langle \Delta I^2 \rangle \cdot e^{-\frac{|\Delta t|}{\tau_c}}$$

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Properties of the current-current correlation function

$C(0) = \langle (\Delta I(t))^2 \rangle > 0$, Mean squared deviation of the current

$$|C(\Delta t)| < C(0), C(\Delta t) = C(-\Delta t), \lim_{\Delta t \rightarrow \infty} C(\Delta t) = 0$$

The Fourier transform of the correlation function:

$t' = t$ variable change

Unit:

$$c(\omega) = \int_{-\infty}^{\infty} dt e^{-i\omega t} C(t) = \int_0^{\infty} dt e^{-i\omega t} C(t) + \int_0^{\infty} dt' e^{i\omega t'} C(t') = 2 \int_0^{\infty} dt C(t) \cos(\omega t) \quad [c(\omega)] = \frac{A^2}{Hz}$$

$c(\omega)$ is real, $c(\omega) = c(-\omega)$ from the cosine formula these properties follow

Inverse transformation: $C(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t} c(\omega)$

From this the mean squared deviation of the current: $\langle (\Delta I(t))^2 \rangle = C(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega c(\omega) = \frac{1}{\pi} \int_0^{\infty} d\omega c(\omega)$

On the other hand, the definition of the spectral density of noise yielded the

$$\langle (\Delta I(t))^2 \rangle = \int_0^{\infty} df S_I(f) = \frac{1}{2\pi} \int_0^{\infty} d\omega S_I(\omega = 2\pi f) \text{ formula for the mean squared deviation of the current.}$$

Comparing the two, we realize that **the spectral density of the current noise is twice the Fourier transform of the current-current correlation function (a more rigorous proof is also available):**

$$S_I(\omega = 2\pi f) = 2c(\omega)$$

The factor two difference is due to the fact that in the experimental definition of S_I we integrate only for positive frequencies, while for $c(\omega)$ we integrate also for negative frequencies. The same considerations apply for the noise of any quantity, not just current.

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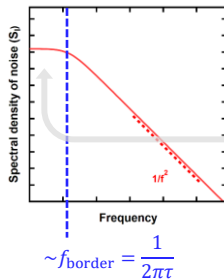
Examples of correlation functions

Exponentially decaying correlation function:

$$C(\Delta t) = C_0 \cdot e^{-\frac{|\Delta t|}{\tau}} = \langle \Delta I^2 \rangle \cdot e^{-\frac{|\Delta t|}{\tau}}$$

$$S(\omega) = 2c(\omega) = 2 \int_0^{\infty} dt e^{-i\omega t} C_0 e^{-\frac{t}{\tau}} + 2 \int_0^{\infty} dt e^{i\omega t} C_0 e^{-\frac{t}{\tau}} = \dots = \frac{4C_0\tau}{1 + \omega^2\tau^2}$$

$$\text{For } \omega \ll 1/\tau: S_0 = 4C_0\tau$$

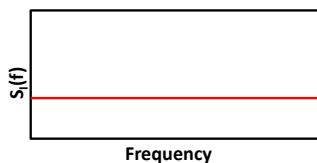


This is the so-called **Lorentzian noise spectrum**. Below the border frequency ($\sim$ inverse correlation time) a frequency independent noise is observed, here it looks "white noise". At higher frequencies the noise power decays with $1/f^2$.

White noise: Dirac-delta correlation function, i.e. current noise values at different times are completely uncorrelated ("zero correlation time"):

$$C(\Delta t) = C_0 \delta(\Delta t)$$

$$S(\omega) = 2c(\omega) = 2 \int_{-\infty}^{\infty} dt e^{-i\omega t} C_0 \delta(t) = 2C_0 = S_0$$



White noise: the spectrum is constant (frequency independent) in the entire frequency range

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How to calculate the noise spectrum in practice by FFT of the measured signal?

Fourier transform of the current deviation: $\Delta i(\omega) = \int_{-\infty}^{\infty} dt \Delta I(t) e^{-i\omega t}$

And the expected value of its absolute value squared for a number of independent measurements:

$$\begin{aligned} \langle |\Delta i(\omega)|^2 \rangle &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt' dt \frac{\langle \Delta I(t) \Delta I(t') \rangle}{c(t'-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' c(\omega') \exp(i\omega'(t'-t))} e^{-i\omega t} e^{i\omega t'} = \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt d\omega' c(\omega') e^{-i(\omega+\omega')t} \underbrace{\int_{-\infty}^{\infty} dt' \frac{e^{i(\omega+\omega')t'}}{2\pi}}_{\delta(\omega+\omega')} = c(\omega) \int_{-\infty}^{\infty} dt \end{aligned}$$

$$\langle |\Delta i(\omega)|^2 \rangle = c(\omega) \cdot \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} dt \quad S_I(\omega) = 2c(\omega) = \lim_{T \rightarrow \infty} \frac{2}{T} \left| \int_{-T/2}^{T/2} dt \Delta I(t) e^{-i\omega t} \right|^2$$

That is, $S_I(\omega)$ is proportional to the expected value of the Fourier transform's absolute value squared, but for an infinite-time Fourier integral the proportionality factor diverges.

What about approximating the above formula for $S_I(\omega)$ by the DFT (or FFT) of the signal as follows? We will check the proper normalization on the next slide.

$$PSD = S_I(\omega) \approx \frac{2}{N\Delta t} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \Delta t \right|^2 = \frac{2\Delta t}{N} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \right|^2$$

This is the practical method of calculating the noise from the data in a measurement

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Normalization of the DFT-based noise formula for various window functions

Calculating the spectral density of noise for a general window function:

$F(t) = N(t) \leftarrow$ Our signal is white noise (frequency independent noise density, time uncorrelated current fluctuation) Of course, it is sufficient to satisfy the white noise condition at the bandwidth corresponding to the sampling rate)

Fourier transform of the signal multiplied by the window function: $n_W(\omega) = \int_{-\infty}^{\infty} W(t) N(t) e^{-i\omega t} dt$

$$\langle |n_W(\omega)|^2 \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt dt' W(t) W(t') \frac{\langle N(t) N(t') \rangle}{c(t'-t) = S_0 \delta(t'-t)/2} e^{-i\omega t} e^{i\omega t'} = \frac{S_0}{2} \int_{-\infty}^{\infty} W^2(t) dt$$

For white noise

This formula is already decent, as it includes the finite measurement time through the window function, and therefore it can be definitely approximated by the discrete sum of the DFT:

$$PSD = S_0 = \frac{2 \langle |n_W(\omega_0)|^2 \rangle}{\int_{-\infty}^{\infty} W^2(t) dt} \approx \frac{2}{\sum_{n=0}^{N-1} W^2(n\Delta t)} \left| \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega n\Delta t} \Delta t \right|^2$$

For a normalized square window $W(t) = 1/T$ is used during the time of the measurement, $\int_{-\infty}^{\infty} W^2(t) dt = 1/T$, but $W(t)$ and Δt also appear in the sum. Collecting all these to the prefactor, really the formula of the previous slide follows:

$$PSD = S_I(\omega) \approx \frac{2\Delta t}{N} \left| \sum_{n=0}^{N-1} F(n\Delta t) e^{-i\omega n\Delta t} \right|^2$$

But this formula is invalid if the DFT is calculated by another window. For a Hanning window the normalization differs by a factor of 1.5, please check if you wish!

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